

Assistance Needs Observed in Human-Human Coassembly

851 annotated assistance events

PHYSICAL SUPPORT — 208/851 (24.4%)

Part out of reach 172/851 (20.2%)

Workpiece stabilization 36/851 (4.2%)

PART SEARCH & TASK DECISIONS — 247/851 (29.0%)

Unable to locate part 124/851 (14.6%)

Uncertain about next step 123/851 (14.5%)

ASSEMBLY ERROR CORRECTION — 301/851 (35.4%)

Incorrect part selection 96/851 (11.3%)

Assembly-order violation 63/851 (7.4%)

Incorrect part orientation 63/851 (7.4%)

Incorrect screwdriver / bit 57/851 (6.7%)

Missed fastener 22/851 (2.6%)

CONVERSATIONAL SUPPORT — 95/851 (11.2%)

Confirmation / affirmation 52/851 (6.1%)

Other conversational support 43/851 (5.1%)

0% 5% 10% 15% 20%

Percentage of All Assistance Events